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ional deformity of the spine that encompasses the curvature of the coronal plane and rotation on the axial axis, with maximum rotational and translational deviations occurring at the apex of the curve.<sup>1</sup> The classification by King et al. was first published in 1983, evaluates the deformity in the coronal plane, describes five types of thoracic curves, and recommends levels of vertebral segments to be included in the arthrodesis.<sup>2</sup>

The Lenke et al. classification (LC) was developed in 2001 and has three components: Six types of curves (1 to 6); the lumbar column modifier (A, B or C); and the sagittal thoracic modifier (-, N or +).<sup>3</sup> The authors emphasize that the corrective surgical strategy for AIS may need to incorporate spinopelvic parameters in the classification to achieve a balanced spine, requiring minimal energy expenditure.<sup>4</sup>

Coonrad et al. reviewed two thousand cases of idiopathic scoliosis and described 21 types of curves. The author indicated 98.7% and 100% for inter-observer and intra-observer reliability, respectively.<sup>5</sup> Kadoury et al. (2012) seek to understand how to classify and quantify three-dimensionally in (3D) the deformities of the spine. They studied 170 patients with AIS, Lenke type 1, when the results showed that multiple representation can potentially be useful for the classification of spinal pathologies in 3D, such as AIS, and serve as a tool to understand the progression of deformities in longitudinal studies.<sup>6</sup> Qiu et al. defined the PUMC classification system in 2005. This method is useful for determining the surgical approach and defining the related fusion levels, and there are 3 main categories such as Type I (single curve), Type II (double curve), and Type III (triple curve) with a total of 13 subtypes.<sup>7</sup>

Comparing the PUMC and Lenke classification systems, Qiu et al. stated that the PUMC classification system is relatively simpler and the inter- and intra-observer variability causes less confusion in surgical planning.<sup>8</sup> In 2018, Garcia et al. created the Three-Dimensional Classification (CT), composed of three components: the first called the quantitative factor, which is divided into three types according to the number of curves; the second, the locator factor, highlights the main curve, and the third, sagittal factor, which globally assesses the sagittal plane.<sup>9</sup>

## OBJECTIVE

Present the modifications of the Modified Three-Dimensional Classification (MTC), and show the results of its application in the classification of radiographic images of the spine, of four hundred and four patients operated on for Adolescent Idiopathic Scoliosis (AIS).

## MATERIAL AND METHOD

Project approved by Scientific Committee. The study had the knowledge free, clarified, and signed by all participants. In 2023, four hundred and four medical records of patients with AIS were studied, all of whom underwent surgical correction using transpedicular fixation in our service, without any specific selection, and they presented common characteristics of the disease in question. All complete and available records have been included. 83 cases that did not meet the requirements were excluded.

Complementary changes were made to the Three-Dimensional Classification, such as the inclusion of Sacral Incidence in the measurement of the sagittal plane. Residual curves are considered to be all those equal to or greater than 10°. The column was divided

## RESULTS

Table 1. Description of the characteristics evaluated in all patients.

Table 2. Description of the scales according to the patients' sex and the results of the association tests.

Table 2 shows that in female patients the frequency of hyper (+) lumbar sagittal factor was statistically higher than in male patients ( $p = 0.033$ ).

According to Table 3, there was no statistically significant association between the patients' color and the evaluated scales ( $p > 0.05$ ).

Table 4 shows that there was a statistically significant average difference in the age of patients according to the thoracic sagittal factor ( $p = 0.009$ ), with the patient with this factor of hypokyphosis being younger than the patient with this normal factor ( $p = 0.007$ ).

Table 5 shows that there was a statistically significant association of all other scales with the quantitative factor ( $p < 0.05$ ), with higher quantitative values being associated with lower expression in the sagittal factors, and patients with factor 2 presented statistically more of locator factor C ( $p < 0.001$ ).

According to Table 6, there was no statistically significant association of the locator factor with the sagittal factors ( $p > 0.05$ ).

|                                        |            |
|----------------------------------------|------------|
|                                        | 353 (87.4) |
|                                        | 51 (12.6)  |
|                                        | 361 (89.4) |
| Black                                  | 20 (5)     |
| Brown                                  | 23 (5.7)   |
| <b>Quantitative factor, n (%)</b>      |            |
| 1                                      | 111 (27.5) |
| 2                                      | 224 (55.4) |
| 3                                      | 69 (17.1)  |
| <b>Locator factor, n (%)</b>           |            |
| A                                      | 12 (3)     |
| B                                      | 266 (65.8) |
| C                                      | 126 (31.2) |
| <b>Thoracic Sagittal Factor, n (%)</b> |            |
| Hipo                                   | 64 (15.8)  |
| Normal                                 | 257 (63.6) |
| Hiper                                  | 83 (20.5)  |
| <b>Lumbar Sagittal Factor, n (%)</b>   |            |
| Hipo                                   | 45 (11.1)  |
| Normal                                 | 281 (69.6) |
| Hiper                                  | 78 (19.3)  |
| <b>Sacral Sagittal Factor, n (%)</b>   |            |
| Hipo                                   | 62 (15.3)  |
| Normal                                 | 315 (78)   |
| Hiper                                  | 27 (6.7)   |

**Table 2.** Description of the scales according to the sex of the patients and the results of the association tests.

| Variable                        | Gender           |                 | p             |
|---------------------------------|------------------|-----------------|---------------|
|                                 | Female           | Male            |               |
| <b>Quantitative Factor</b>      |                  |                 | <b>0.252</b>  |
| 1                               | 101 (28.6)       | 10 (19.6)       |               |
| 2                               | 195 (55.2)       | 29 (56.9)       |               |
| 3                               | 57 (16.1)        | 12 (23.5)       |               |
| <b>Locator Factor</b>           |                  |                 | <b>0.606#</b> |
| A                               | 10 (2.8)         | 2 (3.9)         |               |
| B                               | 230 (65.2)       | 36 (70.6)       |               |
| C                               | 113 (32)         | 13 (25.5)       |               |
| <b>Thoracic Sagittal Factor</b> |                  |                 | <b>0.548</b>  |
| Hipo                            | 54 (15.3)        | 10 (19.6)       |               |
| Normal                          | 224 (63.5)       | 33 (64.7)       |               |
| Hiper                           | 75 (21.2)        | 8 (15.7)        |               |
| <b>Lumbar Sagittal Factor</b>   |                  |                 | <b>0.033</b>  |
| Hipo                            | 39 (11)          | 6 (11.8)        |               |
| Normal                          | 239 (67.7)       | 42 (82.4)       |               |
| Hiper                           | 75 (21.2)        | 3 (5.9)         |               |
| <b>Sacral Sagittal Factor</b>   |                  |                 | <b>0.903#</b> |
| Hipo                            | 55 (15.6)        | 7 (13.7)        |               |
| Normal                          | 274 (77.6)       | 41 (80.4)       |               |
| Hiper                           | 24 (6.8)         | 3 (5.9)         |               |
| <b>Total</b>                    | <b>353 (100)</b> | <b>51 (100)</b> |               |

Chi-square test; # Likelihood ratio test

According to Table 8, there was a statistically significant association of the lumbar sagittal factor with the sacral sagittal factor, with a direct association between the scales; people who were more hypoexpressive in one parameter were more hypoexpressive in the other parameter, and people who were more hyperexpressive in one parameter were more hyperexpressive in the other ( $p < 0.001$ ).

## DISCUSSION

The authors of this article have been using the MTC, therefore, this methodology was compared to that of Lenke (LC)<sup>3</sup>, as it is considered three-dimensional and widely used, which covers only the six most frequent types of the ten described by the Classification discussed in this article.

|                                 |                  |                 |               |
|---------------------------------|------------------|-----------------|---------------|
| 1                               | 97 (26.9)        | 14 (32.6)       |               |
| 2                               | 204 (56.5)       | 20 (46.5)       |               |
| 3                               | 60 (16.6)        | 9 (20.9)        |               |
| <b>Locator Factor</b>           |                  |                 | <b>0.257#</b> |
| A                               | 11 (3)           | 1 (2.3)         |               |
| B                               | 233 (64.5)       | 33 (76.7)       |               |
| C                               | 117 (32.4)       | 9 (20.9)        |               |
| <b>Thoracic Sagittal Factor</b> |                  |                 | <b>0.112</b>  |
| Hipo                            | 59 (16.3)        | 5 (11.6)        |               |
| Normal                          | 233 (64.5)       | 24 (55.8)       |               |
| Hiper                           | 69 (19.1)        | 14 (32.6)       |               |
| <b>Lumbar Sagittal Factor</b>   |                  |                 | <b>0.619#</b> |
| Hipo                            | 42 (11.6)        | 3 (7)           |               |
| Normal                          | 250 (69.3)       | 31 (72.1)       |               |
| Hiper                           | 69 (19.1)        | 9 (20.9)        |               |
| <b>Sacral Sagittal Factor</b>   |                  |                 | <b>0.761#</b> |
| Hipo                            | 57 (15.8)        | 5 (11.6)        |               |
| Normal                          | 280 (77.6)       | 35 (81.4)       |               |
| Hiper                           | 24 (6.6)         | 3 (7)           |               |
| <b>Total</b>                    | <b>361 (100)</b> | <b>43 (100)</b> |               |

Chi-square test; # Likelihood ratio test

**Table 4.** Description of patients' ages according to the scales and results of comparative tests.

| Variable                 | Age (years)  |  |       |
|--------------------------|--------------|--|-------|
|                          | average ± SD |  |       |
| Quantitative Factor      |              |  | 0.875 |
| 1                        | 14.7 ± 2.4   |  |       |
| 2                        | 14.9 ± 2.2   |  |       |
| 3                        | 14.9 ± 1.8   |  |       |
| Locator Factor           |              |  | 0.784 |
| A                        | 15.3 ± 1.9   |  |       |
| B                        | 14.8 ± 2.1   |  |       |
| C                        | 14.8 ± 2.5   |  |       |
| Thoracic Sagittal Factor |              |  | 0.009 |
| Hipo                     | 14.1 ± 1.8   |  |       |
| Normal                   | 15 ± 2.2     |  |       |
| Hiper                    | 14.8 ± 2.4   |  |       |
| Lumbar Sagittal Factor   |              |  | 0.270 |
| Hipo                     | 14.5 ± 1.8   |  |       |
| Normal                   | 14.8 ± 2.2   |  |       |
| Hiper                    | 15.1 ± 2.2   |  |       |
| Sacral Sagittal Factor   |              |  | 0.462 |
| Hipo                     | 14.6 ± 2     |  |       |
| Normal                   | 14.8 ± 2.2   |  |       |
| Hiper                    | 15.2 ± 2.4   |  |       |

ANOVA; & Hipo in average younger than normal ( $p = 0.007$ )

**Table 5.** Description of the other scales according to the quantitative factor and the results of the association tests.

| Variable                        | Quantitative Factor |                  |                 | p                 |
|---------------------------------|---------------------|------------------|-----------------|-------------------|
|                                 | 1                   | 2                | 3               |                   |
| <b>Locator Factor</b>           |                     |                  |                 | <b>&lt;0.001#</b> |
| A                               | 2 (1.8)             | 10 (4.5)         | 0 (0)           |                   |
| B                               | 90 (81.1)           | 116 (51.8)       | 60 (87)         |                   |
| C                               | 19 (17.1)           | 98 (43.8)        | 9 (13)          |                   |
| <b>Thoracic Sagittal Factor</b> |                     |                  |                 | <b>0.024</b>      |
| Hipo                            | 15 (13.5)           | 38 (17)          | 11 (15.9)       |                   |
| Normal                          | 62 (55.9)           | 145 (64.7)       | 50 (72.5)       |                   |
| Hiper                           | 34 (30.6)           | 41 (18.3)        | 8 (11.6)        |                   |
| <b>Lumbar Sagittal Factor</b>   |                     |                  |                 | <b>0.031</b>      |
| Hipo                            | 4 (3.6)             | 30 (13.4)        | 11 (15.9)       |                   |
| Normal                          | 82 (73.9)           | 150 (67)         | 49 (71)         |                   |
| Hiper                           | 25 (22.5)           | 44 (19.6)        | 9 (13)          |                   |
| <b>Sacral Sagittal Factor</b>   |                     |                  |                 | <b>0.026#</b>     |
| Hipo                            | 9 (8.1)             | 41 (18.3)        | 12 (17.4)       |                   |
| Normal                          | 98 (88.3)           | 167 (74.6)       | 50 (72.5)       |                   |
| Hiper                           | 4 (3.6)             | 16 (7.1)         | 7 (10.1)        |                   |
| <b>Total</b>                    | <b>111 (100)</b>    | <b>224 (100)</b> | <b>69 (100)</b> |                   |

Chi-square test; # Likelihood ratio test

|                               |                 |                  |                  |        |
|-------------------------------|-----------------|------------------|------------------|--------|
|                               | )               | 47 (17.7)        | 17 (13.5)        | 0.172# |
|                               | 7)              | 169 (63.5)       | 80 (63.5)        |        |
|                               | 3)              | 50 (18.8)        | 29 (23)          | 0.230# |
| <b>Lumbar Sagittal Factor</b> |                 |                  |                  |        |
| Hipo                          | 0 (0)           | 34 (12.8)        | 11 (8.7)         |        |
| Normal                        | 10 (83.3)       | 185 (69.5)       | 86 (68.3)        |        |
| Hiper                         | 2 (16.7)        | 47 (17.7)        | 29 (23)          |        |
| <b>Sacral Sagittal Factor</b> |                 |                  |                  | 0.251# |
| Hipo                          | 1 (8.3)         | 35 (13.2)        | 26 (20.6)        |        |
| Normal                        | 9 (75)          | 213 (80.1)       | 93 (73.8)        |        |
| Hiper                         | 2 (16.7)        | 18 (6.8)         | 7 (5.6)          |        |
| <b>Total</b>                  | <b>12 (100)</b> | <b>266 (100)</b> | <b>126 (100)</b> |        |

# Likelihood ratio test

**Table 7.** Description of the other scales according to thoracic sagittal factor and the results of the association tests.

| Variable                        | Sacral Sagittal Factor |                  |                 | p                |
|---------------------------------|------------------------|------------------|-----------------|------------------|
|                                 | Hipo                   | Normal           | Hiper           |                  |
| <b>Lumbar Sagittal Factor</b>   |                        |                  |                 | <b>&lt;0.001</b> |
| Hipo                            | 28 (43.8)              | 14 (5.4)         | 3 (3.6)         |                  |
| Normal                          | 35 (54.7)              | 195 (75.9)       | 51 (61.4)       |                  |
| Hiper                           | 1 (1.6)                | 48 (18.7)        | 29 (34.9)       |                  |
| <b>Thoracic Sagittal Factor</b> |                        |                  |                 | 0.189#           |
| Hipo                            | 16 (25)                | 35 (13.6)        | 11 (13.3)       |                  |
| Normal                          | 46 (71.9)              | 203 (79)         | 66 (79.5)       |                  |
| Hiper                           | 2 (3.1)                | 19 (7.4)         | 6 (7.2)         |                  |
| <b>Total</b>                    | <b>64 (100)</b>        | <b>257 (100)</b> | <b>83 (100)</b> |                  |

Chi-square test; # Likelihood ratio test

**Table 8.** Description of the other scales according to lumbar sagittal factor and the results of the association tests.

| Variable                      | Lumbar Sagittal Factor |                  |                 | p                 |
|-------------------------------|------------------------|------------------|-----------------|-------------------|
|                               | Hipo                   | Normal           | Hiper           |                   |
| <b>Sacral Sagittal Factor</b> |                        |                  |                 | <b>&lt;0.001#</b> |
| Hipo                          | 25 (55.6)              | 34 (12.1)        | 3 (3.8)         |                   |
| Normal                        | 20 (44.4)              | 236 (84)         | 59 (75.6)       |                   |
| Hiper                         | 0 (0)                  | 11 (3.9)         | 16 (20.5)       |                   |
| <b>Total</b>                  | <b>45 (100)</b>        | <b>281 (100)</b> | <b>78 (100)</b> |                   |

# Likelihood ratio test

is based on the relationship between the central vertical sacral line and the apex of the lumbar curve; therefore, since types 1 and 2 of this classification do not present a structured curve in the lumbar column, this modifier does not apply. Types 5 and 6 are invariably considered C. Therefore, the Lumbar Modifier is only relevant in types 3 and 4.

The radiological visualization of the T5 vertebra in the lateral view is undoubtedly more difficult due to its overlap with the scapulae, especially in cases of thoracic hypolordosis. Therefore, the option to use T5 as a reference to evaluate the alignment of the sagittal plane, as standardized in the LC, is concerning.

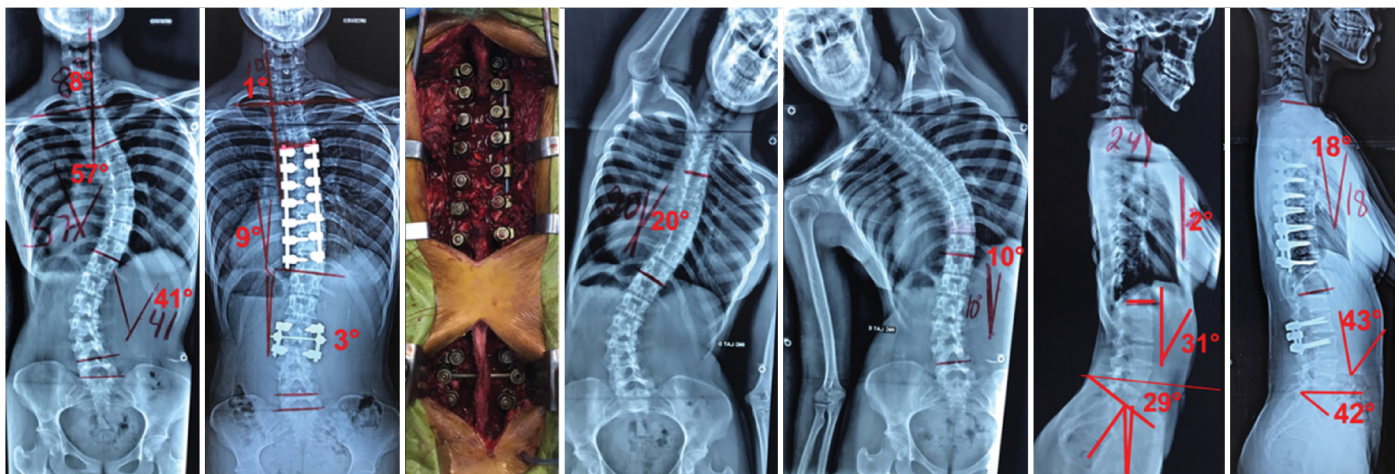
The MTC emphasizes a comprehensive assessment of the sagittal plane, including the measurement of thoracic kyphosis, lumbar lordosis, and sacral incidence, starting the measurement at the terminal vertebrae of the curvatures, which in our experience seems easier and safer.

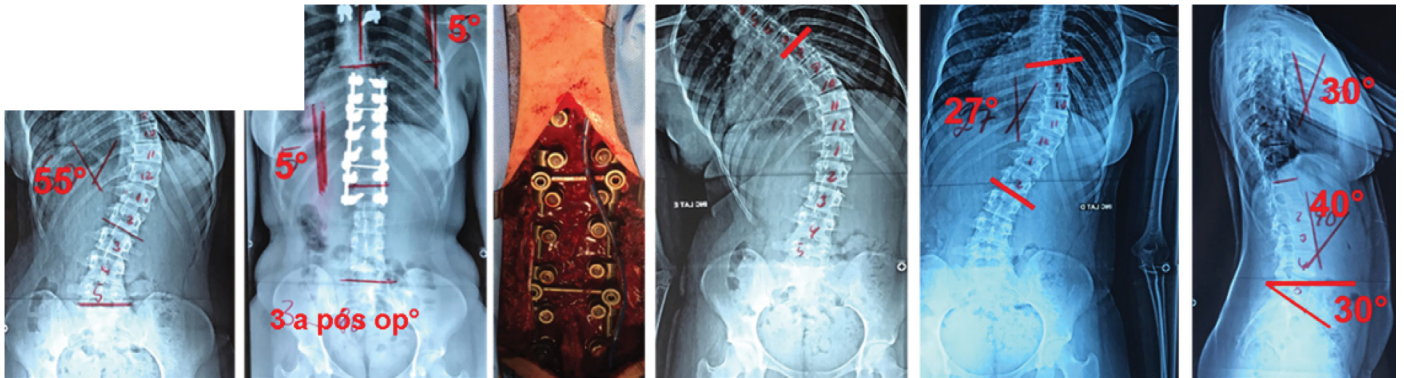
The information about the alignment of the sagittal plane derived only from the measurement of the kyphosis obtained from T5 to T12 (seven vertebrae), remembering that the spine is composed of thirty-three vertebrae and the sagittal plane is composed of four structured curves. Therefore, the use of the LC does not warrant being considered a three-dimensional analysis of spinal deformities. Your assessment of the curves and sagittal balance is quite similar to the classifications of King et al<sup>2</sup> and Coonrad et al<sup>5</sup>, which focus on the coronal plane.

For support in the surgical correction of scoliosis, MTC values the full assessment of the coronal and sagittal planes, getting closer to the Kadoury Classification System<sup>6</sup> and PUMC,<sup>8</sup> focusing on its importance in surgical indications and in the design of scoliosis fixations, see figures 1 to 6.

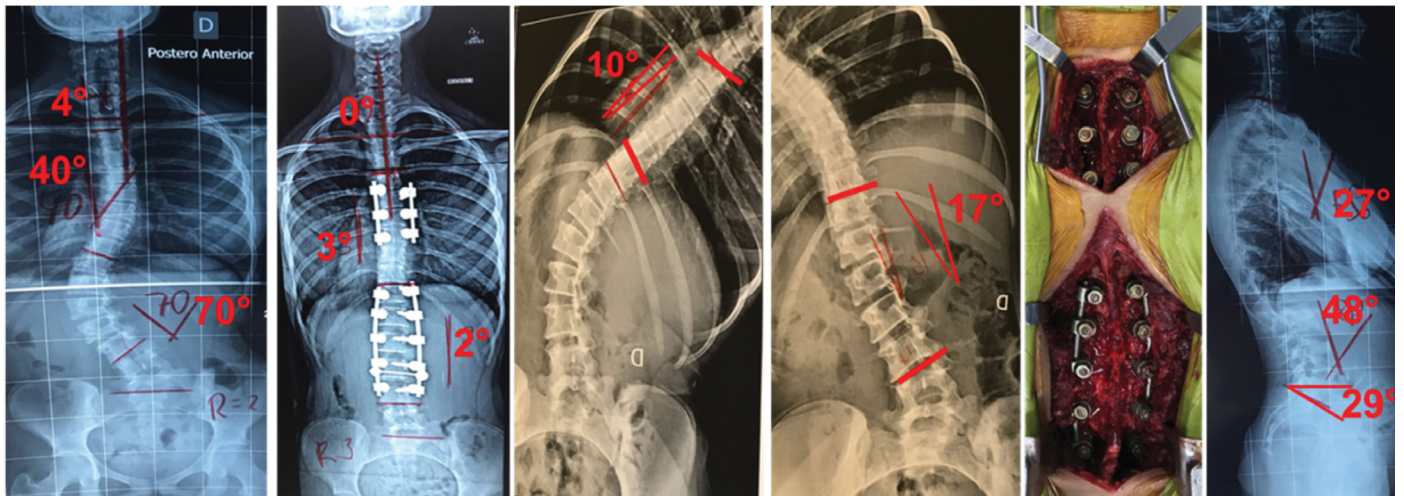
In this study, a higher incidence of double curve scoliosis was observed. Predominance of the main curve in the distal thoracic region. The sagittal plane showed a higher incidence of normal angle. Greater presence of hyperlordosis in females. It is the predominance of hypokyphosis in younger individuals. It can be seen in the result tables that these predominances occurred with significant differences, see tables 1 to 8. Findings that we observe in our daily lives in the treatment of scoliosis.

The importance of considering the fixation of the residual secondary curves, and the appreciation of the approach aims to reduce the trunk imbalance rate, believing and not concluding, that they are the main cause of these complications. Their presence,

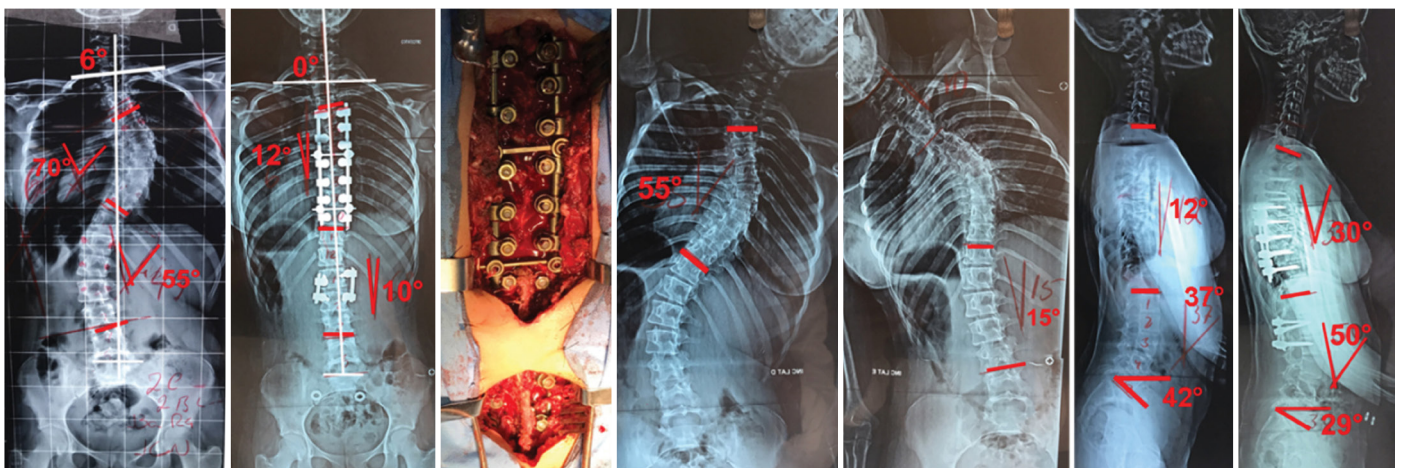




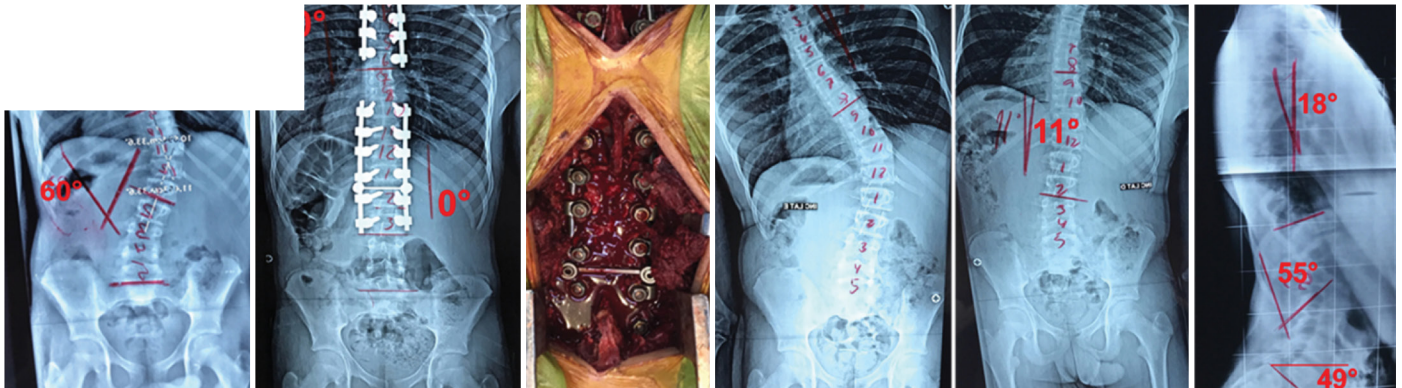
**Figure 2.** 16-year-old female client, classified in the MTC as type 2 BT N N N, residual curves of 22° and 27°, residual curves above 10°, therefore, the two curves were fixed. The SCA corrected itself from 3° to 1°.



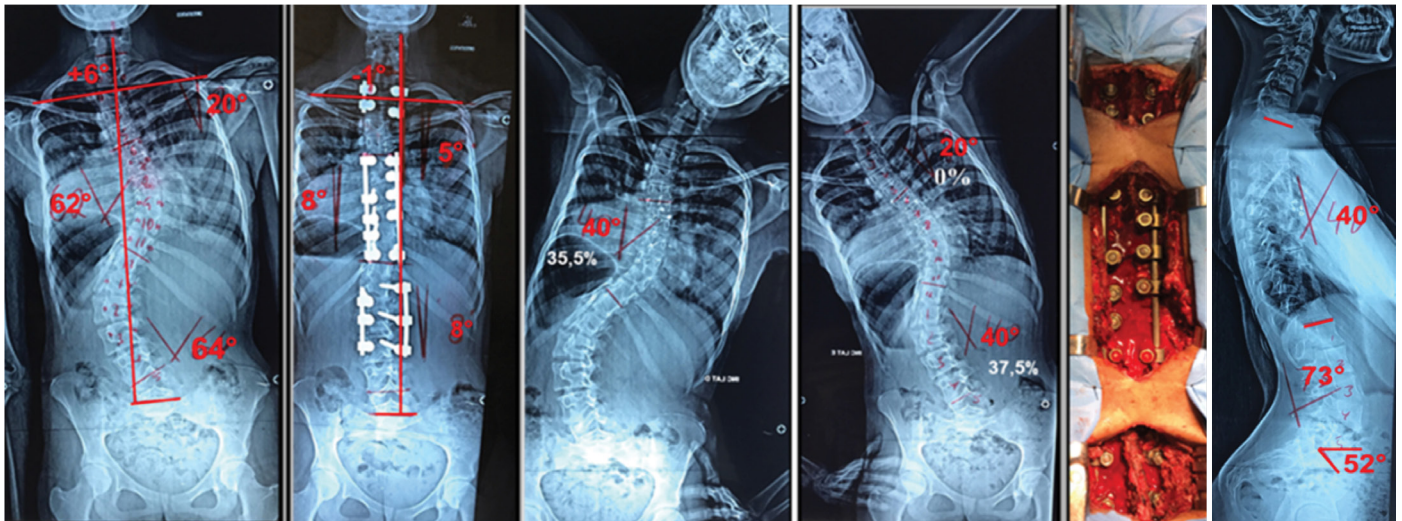
**Figure 3.** 12-year-old girl, female, classified in the MTC as type 2 C N N -, residual curves of 10° and 17°, residual curves above 10°, therefore, surgery was indicated, both curves were addressed. The SCA corrected itself from 4° to 0°.



**Figure 4.** 13-year-old female is classified in the MTC as type 2 BL - N N, residual curves of 55° and 15°, residual curves above 10°, therefore, both curves were addressed. The SCA corrected itself from 6° to 0°.



**Figure 5.** 14-year-old boy, male, is classified in the MTC as type 2 BT - N N, residual curves of 11° and 11°, are above 10°, therefore, both were addressed. The SCA corrected itself from 1° to 0°.



**Figure 6.** 15-year-old female client is classified in the MTC as type 3 C N + +, residual curves of 20°, 40°, and 40° are above 10°, therefore, all three curvatures were addressed. The SCA corrected itself from 6° to -1°.

associated with the existence of one or more neutral vertebrae, can be addressed by Multiple Fixation<sup>12</sup>, with less loss of mobility of the spine. Undoubtedly, the MTC requires further tests and studies in the three-dimensional assessment of the spine.

## CONCLUSIONS

The study of the variables of the MTC applied in the classification of

patients operated on for AIS showed to be of a three-dimensional scope, therefore, it concludes to be a truly three-dimensional classification.

All authors declare no potential conflict of interest related to this article.

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mna. 2018;17(4):275-80.